INFRAMINER AI — EXECUTIVE DECISION REPORT

Multi-Agent Sustainable Resource Sourcing and Resilience Assessment | Generated: 2026-09-23 23:47:26

1. Project Information & Scope

Project Name	Vadhavan Port
Project Type	Port development (container port)
Location Coordinates	Not explicitly documented in source
Timeline / Phases	Phase 1 (initial development with breakwater, reclamation and first terminal) targeted for completion around 2030; Phase 2 (additional terminals and full master-plan expansion) targeted for 2040.
Project Scope Summary	Development of a deep-draft, all-weather container port at Vadhavan, including construction of a 10,140 m offshore breakwater, reclamation, a 1,000 m contiguous berth per terminal (initial Phase 1), multiple container terminals (up to nine 1,000 m berths by 2040), associated quay infrastructure, on-shore facilities on reclaimed land, road and rail access, quarrying and transport of ~32 million tonnes of rock for breakwater, and incorporation of green design and environmental safeguards such as mangrove protection and 300 m shoreline clearance.

Documented Project Constraints:

- No interference with local community
- Independent approaches to the port
- Proximity to an existing fishing harbour and fishermen settlements in the north
- Presence of Shankodhar point in the middle of the sea
- Mangroves and rocky outcrop near the coast
- Minimum 300 m clearance between shoreline and bund to avoid disturbing mangrove flow
- Locating terminal and onshore facilities on reclaimed land
- Minimize impact on adjoining population and fishing community

2. Extracted Material Requirements (12 Items Identified)

Material Name	Quantity	Unit	Specification / Grade	Construction Purpose
Reclamation fill material	177,000,000.0	m3	fill for reclamation work	Reclamation (Phase 1)
Dredged material	7,000,000.0	m3	suitable for reclamation	Reclamation fill
External fill material	170,000,000.0	m3	sourced from external marine borrow pit	Reclamation fill
Murrum fill / burrowed earth	Not available in source	Not available in source	Not available in source	Fill up to +7.0 CD in reclamation
Accropode IITM (concrete armour unit)	Not available in source	Not available in source	Recommended armour size 13 m3 (seaside, trunk) or 11 m3 (roundhead)	Primary armour layer on offshore breakwater
Underlayer rock	Not available in source	kg	2,000-4,000 kg per unit	Under-layer for breakwater
Quarry run (rock)	Not available in source	Not available in source	Size range 1 to 500 kg, coefficient of uniformity (D60/D10) ≥ 3.0	Core material for offshore breakwater
Rock / stone	32,100,000.0	tons	Not available in source	Construction of breakwaters
Rock / stone	16,000,000.0	tons	Quarried rock / stones	Construction of reclamation and shore-protection bund
Geotextile membrane	Not available in source	Not available in source	Not available in source	Rear face of reclamation bund to enhance fill retention
Rock armour (leeside primary)	Not available in source	kg	2,000 - 4,000 kg	Primary armour on leeside slope
Rock armour (lower part of slope)	Not available in source	kg	300 - 1,000 kg	Secondary armour on lower part of leeside slope

3. Candidate Sources & Logistics Overview (63 Matches Evaluated)

Source ID	Source Name	Material	Origin / Location	Lead Distance	Transport Mode
VAD-M01	Offshore Daman Sand Borrow Area (OAMDR Act Reserved Area)	Marine sand / Reclamation fill	Not available in source	N/A	Not available in source
VAD-D01	Vadhavan Port Approach Channel & Navigation Basin Capital Dredging	Dredged material suitable for reclamation fill	Not available in source	N/A	Not available in source
VAD-D01	Vadhavan Port Approach Channel & Navigation Basin Capital Dredging	Dredged material suitable for reclamation fill	Not available in source	N/A	Not available in source
VAD-M01	Offshore Daman Sand Borrow Area (OAMDR Act Reserved Area)	Marine sand / Reclamation fill	Not available in source	N/A	Not available in source
VAD-D01	Vadhavan Port Approach Channel & Navigation Basin Capital Dredging	Dredged material suitable for reclamation fill	Not available in source	N/A	Not available in source
VAD-Q01	Gargaon Quarry (Hillock A) / Patch-1	Basalt rock / Hard murrum	Not available in source	N/A	Not available in source
VAD-Q02	Gargaon Quarry (Hillock B)	Basalt rock / Hard murrum	Not available in source	N/A	Not available in source
VAD-Q03	Khanivade Quarry (Hillock C) / Patch-5	Basalt rock / Hard murrum	Not available in source	N/A	Not available in source
VAD-Q04	Khanivade Quarry (Hillock D)	Basalt rock / Hard murrum	Not available in source	N/A	Not available in source
VAD-Q05	Mahagaon Quarry (Hillock E)	Basalt rock / Hard murrum	Not available in source	N/A	Not available in source
VAD-Q06	Mahagaon Quarry (Hillock F)	Basalt rock / Hard murrum	Not available in source	N/A	Not available in source
VAD-Q07	Nanivali Quarry (Hillock G)	Basalt rock / Hard murrum	Not available in source	N/A	Not available in source

4. Cost, Environmental & Resilience Assessments

• **Cost Estimates:** 63 candidate costings evaluated. Commercial purchase prices are classified as documented or calculated where evidence exists; unstated commercial prices are flagged to avoid fabrication.
• **Environmental Footprint:** 63 evaluations covering quarry extraction footprint, marine dredging guidelines, and overland vs maritime haulage. Where numerical emissions scores are unpublished, neutral baseline assumptions were applied.
• **Supply Chain Resilience:** 63 resilience evaluations factoring multi-quarry redundancy, single-source concessions, and monsoon disruption windows.

5. Multi-Criteria Decision Framework (AHP-TOPSIS)

AHP Criteria Weights: Cost = 0.2685 | Logistics = 0.1213 | Environmental = 0.1899 | Resilience = 0.4203
AHP Consistency Ratio (CR): 0.0262 (CONSISTENT (CR < 0.10))

TOPSIS Evaluated Alternatives (14 Deduplicated Physical Alternatives)

Rank	Source ID	Source Name	Material	TOPSIS Score	Lead Distance	Viability Status
1	VAD-Q02	Gargaon Quarry (Hillock B)	Basalt rock / Hard murrum	**0.8297**	17.0 km	VIABLE
2	VAD-Q04	Khanivade Quarry (Hillock D)	Basalt rock / Hard murrum	**0.8248**	19.0 km	VIABLE
3	VAD-Q05	Mahagaon Quarry (Hillock E)	Basalt rock / Hard murrum	**0.8221**	20.0 km	VIABLE
4	VAD-Q06	Mahagaon Quarry (Hillock F)	Basalt rock / Hard murrum	**0.8221**	20.0 km	VIABLE
5	VAD-Q07	Nanivali Quarry (Hillock G)	Basalt rock / Hard murrum	**0.8079**	25.0 km	VIABLE
6	VAD-Q08	Borsheti Quarry	Basalt stone / Aggregate	**0.8047**	24.0 km	VIABLE
7	VAD-Q09	Kirat Quarry	Basalt stone / Aggregate	**0.8018**	25.0 km	VIABLE
8	VAD-Q10	Nagzari Quarry	Basalt stone / Aggregate / Armour stone	**0.7988**	26.0 km	VIABLE
9	VAD-Q11	Girnoli Quarry	Basalt stone / Aggregate	**0.7926**	28.0 km	VIABLE
10	VAD-Q01	Gargaon Quarry (Hillock A) / Patch-1	Basalt rock / Hard murrum	**0.7697**	17.0 km	VIABLE
11	VAD-Q03	Khanivade Quarry (Hillock C) / Patch-5	Basalt rock / Hard murrum	**0.7661**	19.0 km	VIABLE
12	VAD-D01	Vadhavan Port Approach Channel & Navigation Basin Capital Dredging	Dredged material suitable for reclamation fill	**0.7434**	5.0 km	VIABLE
13	VAD-M02	Vadhavan Port Concrete Element Casting Yard	Accropode II concrete armour units (11 cum and 13 cum units)	**0.6761**	3.0 km	VIABLE
14	VAD-M01	Offshore Daman Sand Borrow Area (OAMDR Act Reserved Area)	Marine sand / Reclamation fill	**0.0000**	55.0 km	VIABLE

6. Material-Wise Final Sourcing & Resilience Plan

Material	Primary Source Recommendation	Score	Fallback Contingency Sources	Resilience & Viability Status
Basalt rock / Hard murrum	Gargaon Quarry (Hillock B)	0.8297	• Khanivade Quarry (Hillock D) (VAD-Q04) • • Mahagaon Quarry (Hillock E) (VAD-Q05) • • Mahagaon Quarry (Hillock F) (VAD-Q06) • • Nanivali Quarry (Hillock G) (VAD-Q07) • • Gargaon Quarry (Hillock A) / Patch-1 (VAD-Q01) • • Khanivade Quarry (Hillock C) / Patch-5 (VAD-Q03)	RESILIENT (High redundancy)
Basalt stone / Aggregate	Borsheti Quarry	0.8047	• Kirat Quarry (VAD-Q09) • • Girnoli Quarry (VAD-Q11)	RESILIENT (High redundancy)
Basalt stone / Aggregate / Armour stone	Nagzari Quarry	0.7988	None (Single-source dependency)	**SINGLE-SOURCE DEPENDENCY**
Dredged material suitable for reclamation fill	Vadhavan Port Approach Channel & Navigation Basin Capital Dredging	0.7434	None (Single-source dependency)	**SINGLE-SOURCE DEPENDENCY**
Accropode II concrete armour units (11 cum and 13 cum units)	Vadhavan Port Concrete Element Casting Yard	0.6761	None (Single-source dependency)	**SINGLE-SOURCE DEPENDENCY**
Marine sand / Reclamation fill	**No viable source identified**	N/A	None (Single-source dependency)	**NO VIABLE SOURCE**

7. Supply Chain Disruption & Resilience Narrative

Resilience is weighted at 42.0% in the multi-criteria decision framework, reflecting the critical need for supply continuity in large-scale infrastructure projects. Primary selection 'Gargaon Quarry (Hillock B)' (Basalt rock / Hard murrum) achieves the highest overall TOPSIS score (0.8297), optimizing the trade-off across cost, logistics, environmental footprint, and resilience. 10 viable fallback alternative(s) were identified ('Khanivade Quarry (Hillock D)', 'Mahagaon Quarry (Hillock E)', 'Mahagaon Quarry (Hillock F)'). These alternatives provide critical operational buffering against supply chain disruptions, such as seasonal monsoon closures, transport corridor failures, or capacity constraints. Notice: Material 'Basalt stone / Aggregate / Armour stone' exhibits single-source dependency. A proactive risk mitigation strategy is recommended. Notice: Material 'Dredged material suitable for reclamation fill' exhibits single-source dependency. A proactive risk mitigation strategy is recommended. Notice: Material 'Accropode II concrete armour units (11 cum and 13 cum units)' exhibits single-source dependency. A proactive risk mitigation strategy is recommended. Notice: Material 'Marine sand / Reclamation fill' has no viable source identified under multi-criteria threshold. Dedicated procurement action required.

8. Evidence, Provenance & Methodological Audit

• **DPR Evidence Grounding:** Project identity, breakwater structures, and material specifications extracted strictly from indexed DPR pages.
• **Quarry & Material Sourcing Provenance:** Source names, lead distances, and reserve volumes derived from state mining department surveys, CIDCO/MMRDA regional master plans, and environmental clearance documents.
• **Cost & Commercial Pricing Audit:** Documented tariff rates and freight estimates apply standard ministry schedule of rates (SoR). Where commercial unit prices are unpublished, neutral placeholders prevent numerical fabrication.
• **Non-Viability & Threshold Rule:** Alternatives achieving TOPSIS closeness coefficient < 0.30 are formally rejected as non-viable to prevent unfeasible procurement commitments.

9. Operational Limitations & Governance Notice

NOTICE: InfraMiner AI is an automated multi-criteria decision-support system. Sourcing evaluations tagged as PROVISIONAL_INDICATIVE reflect constraints in publicly disclosed DPR documentation. Formal competitive bidding, physical quarry core sampling, and statutory coastal regulatory approvals (CRZ/MoEFCC) are mandatory prior to financial commitment.